



Financial Institutions & Markets- ENT 309

PPT #3, IITD, Summer, 2026

Agenda

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- How to Bond Prices change?
 - With YTM
 - With Maturity
 - With Coupon Rate
- Duration

How do Bond Prices
change?



YTM for a Bond

- Face or Par Value: Rs 1000
- Coupon: 10%
- Years to Maturity: 10 years.
- Price of the Bond Today=1000
- YTM= r = Discount Rate=?

$$P_0 = \sum_{t=1}^T \frac{C}{(1+r)^t} + \frac{F}{(1+r)^T}$$

Diagram illustrating the Bond Value formula with labels:

- BOND VALUE: P_0
- Interest Payment (\$): C
- Par Value: F
- Interest Rate: r
- Number of Periods: T



YTM Formula

- Quick & Dirty Way: Map CF(cash flows) and calculate IRR.
- Proper Way: Calculate YTM using Yield.
- **=Yield(settlement, maturity, rate, pr, redemption, frequency, [basis])**
 - Settlement: Date of purchase
 - Maturity: date of maturity/expiry
 - Rate: coupon rate
 - Pr: the bond's price per face value of 100
 - Redemption: the bond's redemption value per face value of 100
 - Frequency: Number of coupon payments per year
 - Basis: optional, Day count convention. Choose 1 for actual.
 - Answer to be in %



YTM Formula

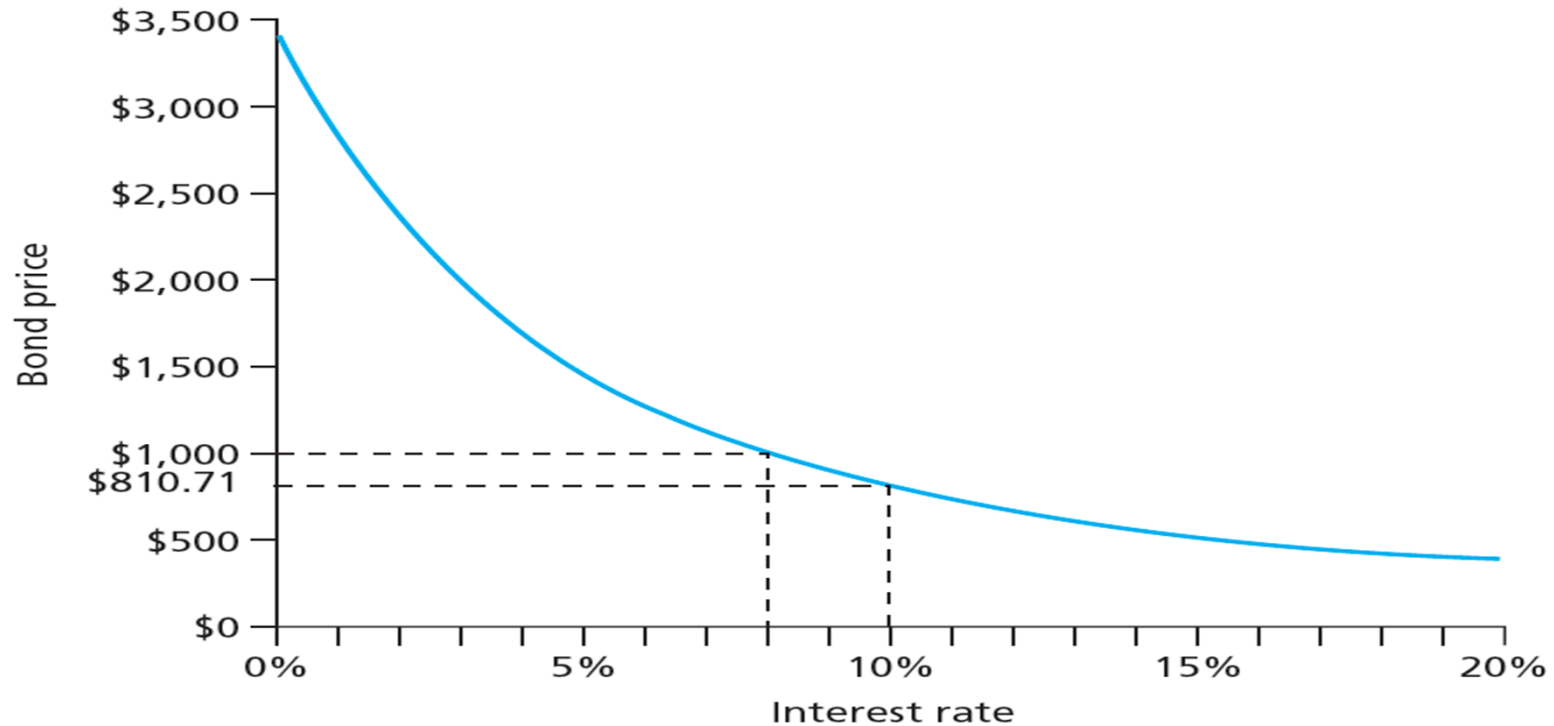
- Date of purchase: 1 Jan 2026
- Date of maturity: 1 Jan 2036
- Coupon: 10%
- Purchase Price: 800
- Face Value or Redemption Value: 1000
- Coupon payment: Semi-annually or half-yearly
- Calculate YTM. IRR
- Change Date of Maturity to 15th Jan 2036 and recalculate.

Change in Bond Price



- What happens to the Price of Bond as YTM rises/falls?
- Inversely proportional
- IF SAME CHANGE IN YTM FOR TWO BONDS WITH DIFFERENT COUPON RATES, PRICE OF WHICH BOND WILL CHANGE MORE?
- IF SAME CHANGE IN YTM FOR TWO BONDS WITH DIFFERENT MATURITY, PRICE OF WHICH BOND WILL CHANGE MORE?

Bond Price Vs Int Rate





Bond Prices at Diff Int Rates

Time to Maturity	Bond Price at Given Market Interest Rate				
	2%	4%	6%	8%	10%
1 year	\$1,059.11	\$1,038.83	\$1,019.13	\$1,000.00	\$981.41
10 years	1,541.37	1,327.03	1,148.77	1,000.00	875.38
20 years	1,985.04	1,547.11	1,231.15	1,000.00	828.41
30 years	2,348.65	1,695.22	1,276.76	1,000.00	810.71

- What is the coupon rate?
- How does the price change with maturity, other things being constant?
- What is the frequency of coupon payment- Yearly or Semi-Annual?



Bond Prices at Diff Int Rates

Par	1000
Coupon	8%
YTM	2%
Price	1059.11

Option 1	Annual Coupon Payout		
	Cash Flow	Disc Factor	Price Today
	1080	0.980	1058.824

Option 1	Semi-Annual Coupon Payout		
	40	0.990	39.604
	1040	0.980	1019.508
	TOTAL		1059.112

- Frequency of coupon payment- Semi-Annual.



Price Change For Diff Maturity

	PV of an 8% bond. Rs.1000 par value. Different maturities							
Term to Maturity (years)	1	1	10	10	20	20	30	30
Discount rate (YTM)	7%	10%	7%	10%	7%	10%	7%	10%
Price of bond	₹ 1,009	₹ 982	₹ 1,070	₹ 877	₹ 1,106	₹ 830	₹ 1,124	₹ 811
% change in total value: YTM changes from 7% to 10%		-2.73%		-18.05%		-24.98%		-27.81%

- Bond Prices moves inversely to Interest Rates
- Longer Maturity bonds respond more strongly to change in interest rate



Price Change For Diff Maturity

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% change in total value: YTM changes from 7% to 10%		-2.73%	22%	-18.05%		-24.98%		-27.81%

- Price changes are greater when interest rates fall than when they rise:
Asymmetry



Price Change: Diff Coupons

	PV of an 20 year bond. Rs.1000 par value. Different coupons							
Coupon Rates	0%	0%	3%	3%	8%	8%	12%	12%
Discount rate (YTM)	7%	10%	7%	10%	7%	10%	7%	10%
Price of bond	₹ 258	₹ 149	₹ 576	₹ 404	₹ 1,106	₹ 830	₹ 1,530	₹ 1,170
% change in total value: YTM changes from 7% to 10%		-42.5%		-29.9%		-25.0%		-23.5%

- Lower Coupon Bonds React More Strongly To Change in Interest Rates

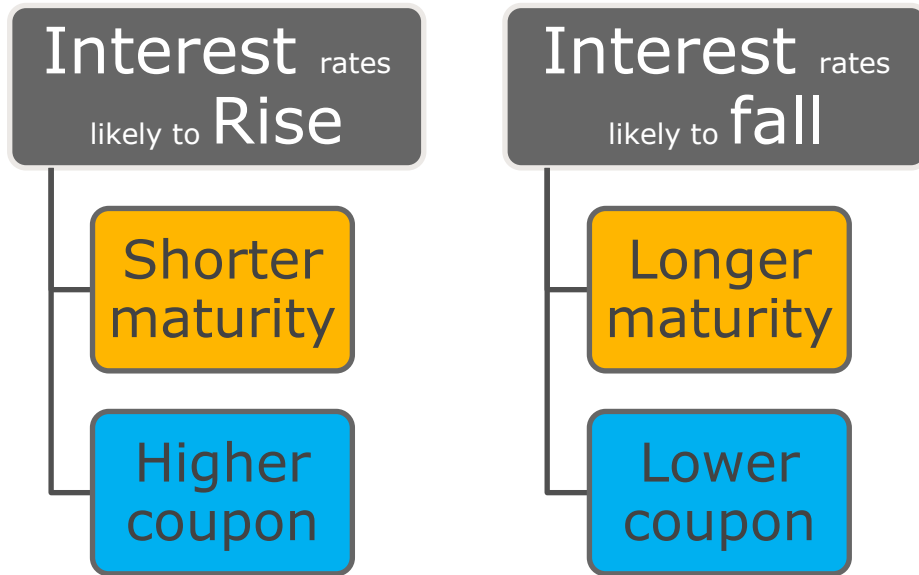
Bond Prices: Summary



- Bond Prices moves inversely to Interest Rates
- Longer Maturity bonds respond more strongly to change in interest rate
- Lower Coupon Bonds React More Strongly To Change in Interest Rates
- Price changes are greater when interest rates fall than when they rise: **Asymmetry**



Implications For Fund Manager



Which bond to select?

	Maturity	Coupon
Bond A	10 years	9%
Bond B	5 years	6%

- Difficult to say from the given data. We need a new concept.
- This concept is called Duration

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Duration



Duration-Pre

- Suppose that we have two bonds, **identical except for their term to maturity**. Both bonds have coupon rates of 10% paid annually, and face values of Rs.1,000. Your current YTM is 10%. Bond 1 has 5 years to maturity while Bond 2 has 10 years to maturity.
- Now, if your YTM drops to 8%, which bond will increase the most in value?
- Bond 1 will be worth Rs.1,079.85, and Bond 2 will be worth Rs.1,134.20
- Bond 2 wins because of its longer maturity
- Note that had rates risen instead, then Bond 1 would lose less than Bond 2 so Bond 1 would be favored.



Duration-Pre

- Now, take a case where both the bonds have 5 years to maturity, but Bond 3 has a coupon rate of 7% and Bond 4 has a coupon rate of 10%.
- With your YTM at 10%, Bond 3 is worth Rs.886.28, and Bond 4 is worth Rs.1,000.
- If your YTM drops to 8%, which bond will increase more in value?
- Bond 3 will be worth Rs.960.73 (an increase of 8.40%), and Bond 4 will be worth Rs.1,079.85 (an increase of 7.99%).
- So, Bond 3, with the lower coupon rate wins



Duration Pre

- Finally, take a Bond 5 which has a maturity of 5 years and a 5% coupon, while Bond 6 has a maturity of 10 years and a 10% coupon?
- Now we have a conflict. Bond 5 has a lower coupon rate, but Bond 6 has a longer maturity. How do we know which one will change more in price when rates drop to 8%?
- First, note that at 10% Bond 5 is worth Rs.810.46 and Bond 6 is worth Rs.1,000.
- At 8%, Bond 5 is worth Rs.880.22 (an 8.6% increase), and Bond 6 is worth Rs.1,134.20 (and increase of 13.42%).
- Bond 6 wins because it has a longer **DURATION**.



Duration- D

Weight associated with the cash flow made at time t (denoted by CF_t)

$$w_t = \frac{CF_t / (1 + y)^t}{\text{Bond price}}$$

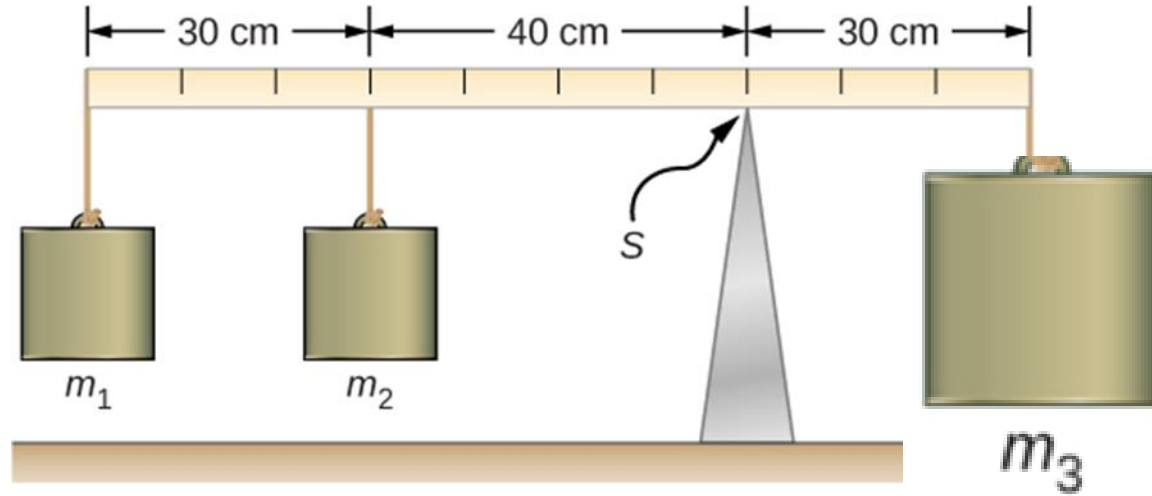
- $CF \rightarrow$ Cash Flow
- $y \rightarrow$ YTM/ Discount Rate
- $t \rightarrow$ time period

Macaulay's Duration (we will refer to it as Duration for practical reasons)

$$D = \sum_{t=1}^T t \times w_t$$

- DURATION IS A MEASURE OF THE "EFFECTIVE MATURITY" OF THE BOND.
- Another useful concept is Modified Duration.

Equivalence: Duration & Centre of Mass



Centre of Mass of a system of Particles

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

Duration

$$D = \frac{\sum t \cdot \frac{CF_t}{(1+y)^t}}{P}$$

Duration Calculation



Let's calculate the durations of Bond 5 and Bond 6 from the last example, using a 10% required return:

$$D_{\text{Bond 5}} = \frac{\frac{50}{1.10^1} \times 1 + \frac{50}{1.10^2} \times 2 + \frac{50}{1.10^3} \times 3 + \frac{50}{1.10^4} \times 4 + \frac{1050}{1.10^5} \times 5}{810.46} = 4.49$$

So, Bond 5 has a duration of **4.49 years**. Similar calculations will show that Bond 6 has a duration of **6.76 years**.

The longer duration is the reason that Bond 6 outperformed Bond 5 when rates fell to 8%.



Duration: Properties

- A longer term to maturity increases duration, all other things being equal.
- For coupon-bearing bonds, duration is always less than term to maturity.
- For zero-coupon bonds it is exactly the same as term to maturity.
- Lower coupon rates lead to longer durations, all other things being equal.
- Higher yields lead to shorter durations.



Macaulay Duration: Uses

- Extremely useful for managing a bond portfolio
- When interest rates are expected to move, a portfolio manager should change the duration of the portfolio to take maximum advantage (or limit damage) of the change.
- Immunization is another technique which fund managers use. You can read up if keen to know more.



Duration: Calculation

- Three year 8% Rs.1000 bond. Prevailing interest rate (YTM) is 10%
- Three year zero coupon Rs. 1000 bond



Duration: Calculation

Interest Rate	10%
Coupon	8%
Maturity (years)	3
Face Value	1000

$$w_t = \frac{CF_t / (1 + y)^t}{\text{Bond price}}$$

Time Until Payment (years)	Payout	Discounted CF	Weight= Disc CF / Price	Duration Components = wt. * t
1	80	72.73	0.077	0.07653384
2	80	66.12	0.070	0.13915244
3	1080	811.42	0.854	2.56166983
PRICE		950.26	1.000	
Duration				2.777

Rate of Return Vs. Interest Rate

- Rate of Return on holding and trading a bond. Capital gains/loss + coupon payouts.
- Interest Rate = YTM



Thank You!